CENG381 Stochastic Processes

Gambler's Ruin - Answer Key 1

Q1 Solution

a) Recurrence and boundary conditions:

At each step from state k ($1 <= k <= N-1$), the gambler moves to $k+1$ (prob $p = 1/2$) or $k-1$ (prob $q = 1/2$). Conditioning on the first step:

$$R(k) = (1/2)*R(k+1) + (1/2)*R(k-1) \quad \text{for } k = 1, 2, ..., N-1$$

Boundary conditions:

$R(0) = 0$ [gambler is ruined, probability of winning = 0]

$R(N) = 1$ [gambler has won all N dollars, probability of winning = 1]

b) Solution for the unbiased case:

Rewrite the recurrence as:

$2*R(k) = R(k+1) + R(k-1)$

$R(k+1) - R(k) = R(k) - R(k-1)$ (constant differences)

So R(k) is linear in k:

$R(k) = A + B*k$

Applying boundary conditions:

$R(0) = A = 0 \implies A = 0$

$R(N) = B*N = 1 \implies B = 1/N$

$$R(k) = k/N \quad \text{for } k = 0, 1, ..., N.$$

c) Result for k=3, N=10:

$R(3) = 3/10 = 0.30$

Interpretation: Starting with 3 dollars against an opponent with 7 dollars in a fair game, the gambler has only a 30% chance of winning all 10 dollars before going broke. The opponent, starting with more capital, has a 70% chance of winning. This illustrates how having less capital puts a gambler at a serious disadvantage, even in a fair game.

Q2 Solution

a) Recurrence and characteristic equation:

$R(k) = (1/3)*R(k+1) + (2/3)*R(k-1)$

Rearranging: $(1/3)*R(k+1) - R(k) + (2/3)*R(k-1) = 0$

Multiply through by 3: $R(k+1) - 3*R(k) + 2*R(k-1) = 0$

Try solution $R(k) = r^k$. Substituting:

$r^{(k+1)} - 3*r^k + 2*r^{(k-1)} = 0$

Divide by $r^{(k-1)}$: $r^2 - 3r + 2 = 0$

$(r - 1)(r - 2) = 0 \Rightarrow r_1 = 1, r_2 = 2$

Note: $r_2 = 2 = q/p = (2/3)/(1/3)$. This matches the formula $r_2 = q/p$.

b) General solution and boundary conditions:

General solution: $R(k) = A * 1^k + B * (q/p)^k = A + B * 2^k$

Apply $R(0) = 0$: $A + B*2^0 = 0 \Rightarrow A + B = 0 \Rightarrow A = -B$

Apply $R(N) = R(6) = 1$: $A + B*2^6 = 1$

$-B + 64*B = 1 \Rightarrow 63*B = 1 \Rightarrow B = 1/63$

$A = -1/63$

$R(k) = (-1/63) + (1/63)*2^k = (2^k - 1) / 63$

c) R(2) and comparison with unbiased case:

$R(2) = (2^2 - 1) / 63 = (4 - 1) / 63 = 3/63 = 1/21 \sim 0.048$

Unbiased value would be: $k/N = 2/6 = 1/3 \sim 0.333$

$R(2) = 1/21 << 1/3$: the biased game is MUCH worse for the gambler.

Reason: with $p = 1/3 < q = 2/3$, each bet is unfavorable. The gambler loses money on average each step, making ruin nearly certain when starting with only 2 out of 6 dollars. The unfavorable odds compound the disadvantage of having less capital.

Q3 Solution

a) Recurrence for $f(k)$:

Conditioning on the first step (each direction with prob 1/2):

$$f(k) = 1 + (1/2) \cdot f(k+1) + (1/2) \cdot f(k-1) \quad \text{for } k = 1, \ldots, N-1$$

[We take 1 step, then continue from k+1 or k-1.]

Boundary conditions:

$f(0) = 0$ [already absorbed at 0, game over]

$f(N) = 0$ [already absorbed at N, game over]

b) Solve the recurrence:

Rearranging: $(1/2) \cdot f(k+1) - f(k) + (1/2) \cdot f(k-1) = -1$

Multiply by 2: $f(k+1) - 2 \cdot f(k) + f(k-1) = -2$

Homogeneous equation: $f(k+1) - 2 \cdot f(k) + f(k-1) = 0$

Characteristic equation: $r^2 - 2r + 1 = (r-1)^2 = 0$

Repeated root $r = 1$ => homogeneous solution: $f_h(k) = A + B \cdot k$

Particular solution for -2 on the right side:

Try $f_p(k) = C \cdot k^2$ (since k^0 and k^1 already in homogeneous part)

Substitute: $C \cdot (k+1)^2 - 2C \cdot k^2 + C \cdot (k-1)^2 = -2$

$C \cdot (k^2+2k+1 - 2k^2 + k^2-2k+1) = -2$

$C \cdot 2 = -2$ => $C = -1$

$f_p(k) = -k^2$

General solution: $f(k) = A + B \cdot k - k^2$

Apply $f(0) = 0$: $A = 0$

Apply $f(N) = 0$: $B \cdot N - N^2 = 0$ => $B = N$

$$f(k) = N \cdot k - k^2 = k \cdot (N - k) \quad \text{(OK)}$$

c) Values for N=8:

$f(k) = k \cdot (8 - k)$

f(2) = 2*(8-2) = 2*6 = 12 steps

f(4) = 4*(8-4) = 4*4 = 16 steps (maximum: k = N/2)

f(6) = 6*(8-6) = 6*2 = 12 steps

The game lasts longest on average when the gambler starts with $k = N/2 = 4$ dollars (equal capital on both sides). By symmetry, $f(k) = f(N-k)$, and the maximum expected duration is $(N/2)^2 = 16$ steps when $N = 8$.